

Commissioning, at *hospital scale*

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CONTACT

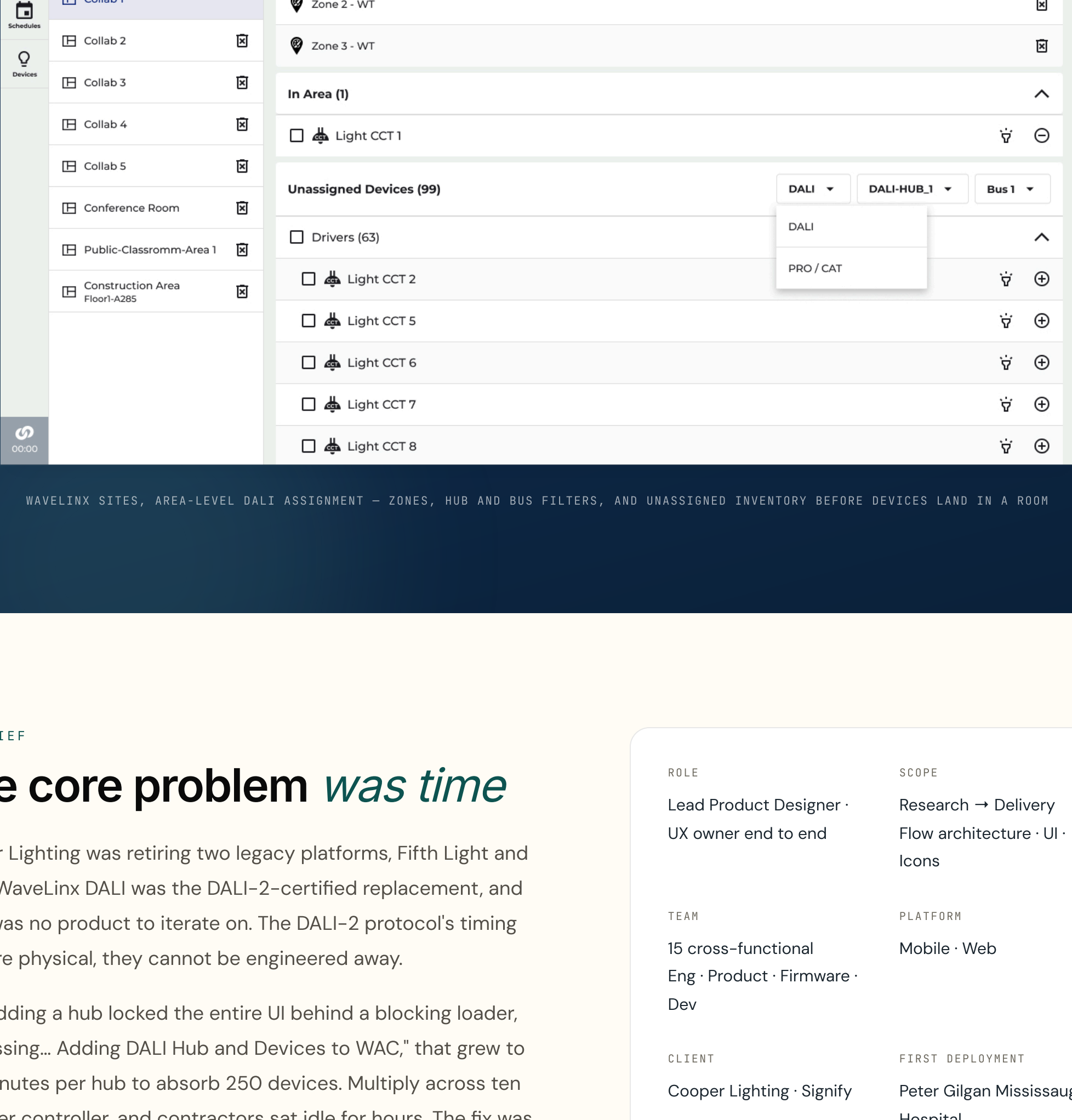


CASE STUDY DALI-2 Lighting System

An industry-first commissioning experience for DALI-2 Lighting System, a protocol-constrained commercial IoT product taken from zero to shipped across hardware, firmware, and field reality at once. First deployment: the largest DALI-2 install in North America.

Read the full case study →

See the commissioning flow



WAVELINX SITES, AREA-LEVEL DALI ASSIGNMENT — ZONES, HUB AND BUS FILTERS, AND UNASSIGNED INVENTORY BEFORE DEVICES LAND IN A ROOM

THE BRIEF

The core problem *was time*

Cooper Lighting was retiring two legacy platforms, Fifth Light and iLight. WaveLinx DALI was the DALI-2-certified replacement, and there was no product to iterate on. The DALI-2 protocol's timing rules are physical, they cannot be engineered away.

In V1, adding a hub locked the entire UI behind a blocking loader, "Processing... Adding DALI Hub and Devices to WAC," that grew to 27.5 minutes per hub to absorb 250 devices. Multiply across ten hubs per controller, and contractors sat idle for hours. The fix was never a longer timeout, it was designing around the wait.

ROLE	SCOPE
Lead Product Designer · UX owner end to end	Research → Delivery Flow architecture · UI · Icons
TEAM	PLATFORM
15 cross-functional Eng · Product · Firmware · Dev	Mobile · Web
CLIENT	FIRST DEPLOYMENT
Cooper Lighting · Signify	Peter Gilgan Mississauga Hospital

CHALLENGE

A protocol that *couldn't change*

Three constraints shaped every screen. None could be removed, the UX had to absorb each one rather than wish it away.

01

A blocking import, by design

Discovery triggered a full scan of all four buses. No navigation, no area creation, no parallel work, no progress, until it finished.

02

Built for 30-50 devices

Every screen, device trees, area lists, the operate floor plan, assumed a handful of wireless devices. DALI-2 brought thousands per floor.

03

Make-or-break, not polish

Channel partners and facilities teams were judging whether WaveLinx DALI could credibly replace their tools. A blocking experience at hospital scale was not acceptable.

THE SCALE

The numbers I was *designing for*

DALI-2's physical limits define the whole problem space. Every interaction had to hold from a single bus to a full hospital floor.

10,000+

Devices per hospital floor

24

Buses per floor

6

DALI hubs per panel

128

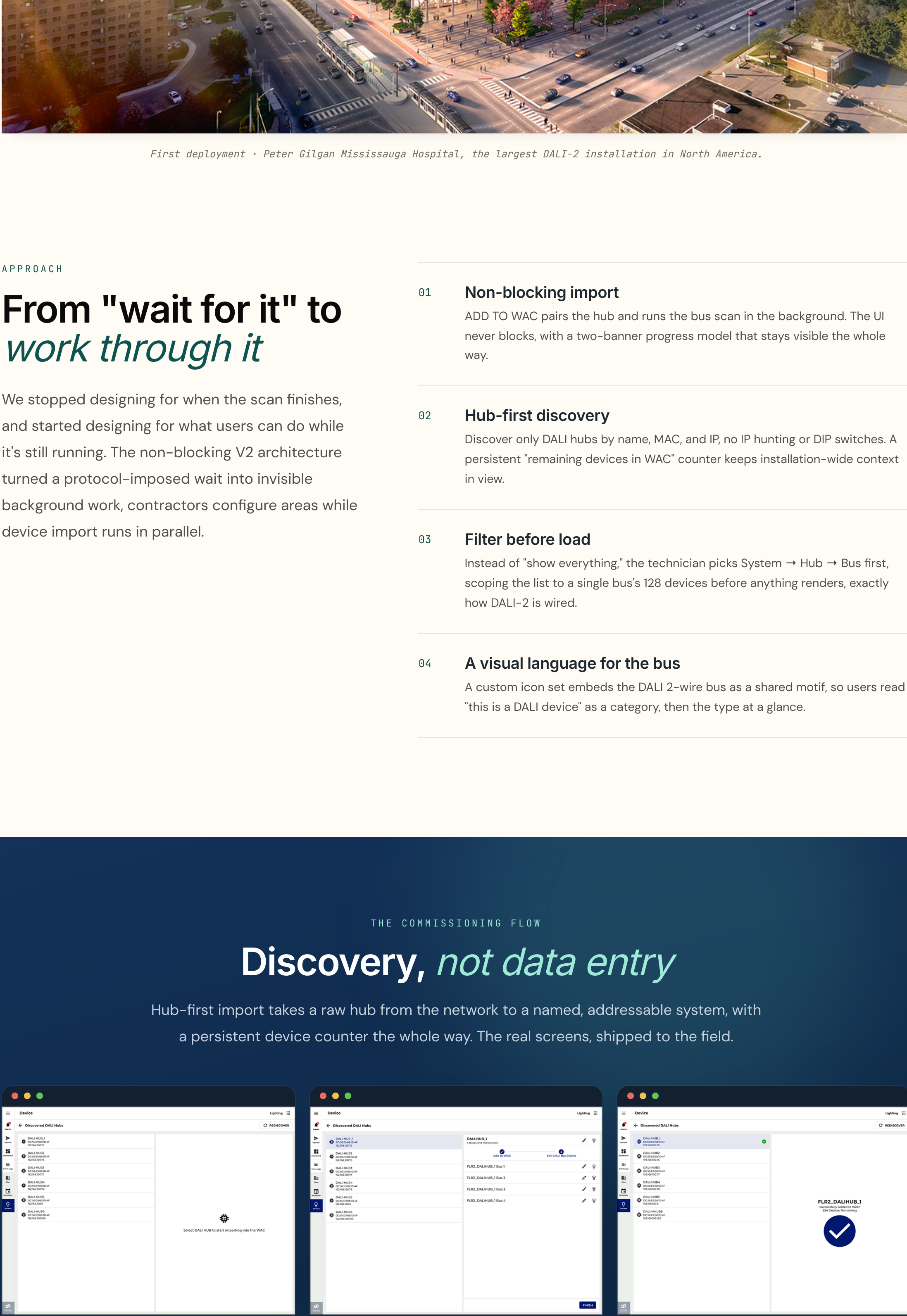
Fixtures per bus

1,000 ft

Max bus length

250

Devices per hub (per scan)



First deployment · Peter Gilgan Mississauga Hospital, the largest DALI-2 installation in North America.

APPROACH

From "wait for it" to *work through it*

We stopped designing for when the scan finishes, and started designing for what users can do while it's still running. The non-blocking V2 architecture turned a protocol-imposed wait into invisible background work, contractors configure areas while device import runs in parallel.

01

Non-blocking import

ADD TO WAC pairs the hub and runs the bus scan in the background. The UI never blocks, with a two-banner progress model that stays visible the whole way.

02

Hub-first discovery

Discover only DALI hubs by name, MAC, and IP, no IP hunting or DIP switches. A persistent "remaining devices in WAC" counter keeps installation-wide context in view.

03

Filter before load

Instead of "show everything," the technician picks System → Hub → Bus first, scoping the list to a single bus's 128 devices before anything renders, exactly how DALI-2 is wired.

04

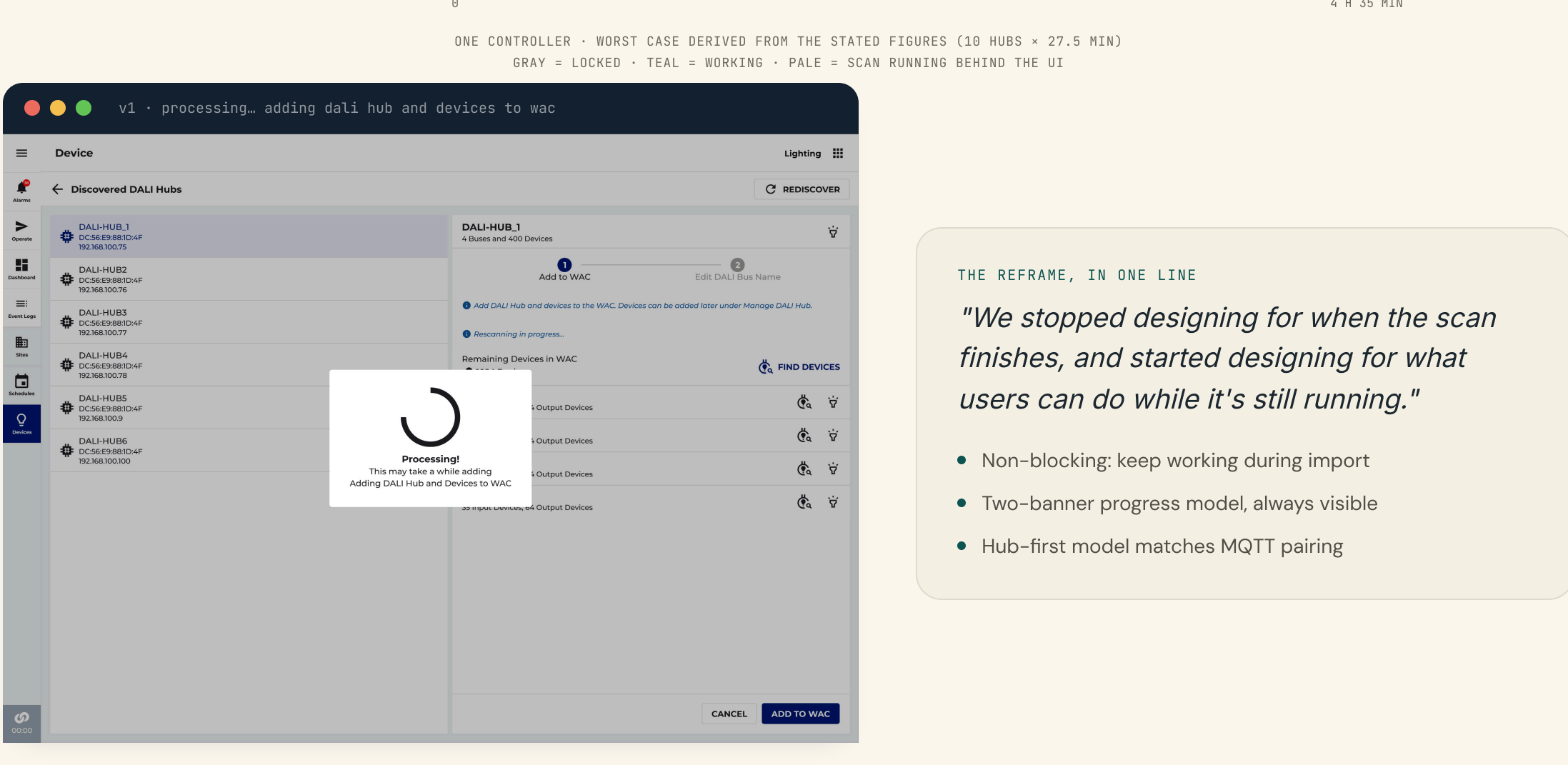
A visual language for the bus

A custom icon set embeds the DALI 2-wire bus as a shared motif, so users read "this is a DALI device" as a category, then the type at a glance.

THE COMMISSIONING FLOW

Discovery, *not data entry*

Hub-first import takes a raw hub from the network to a named, addressable system, with a persistent device counter the whole way. The real screens, shipped to the field.



STEP 1 · DISCOVER, EVERY HUB ON THE NETWORK

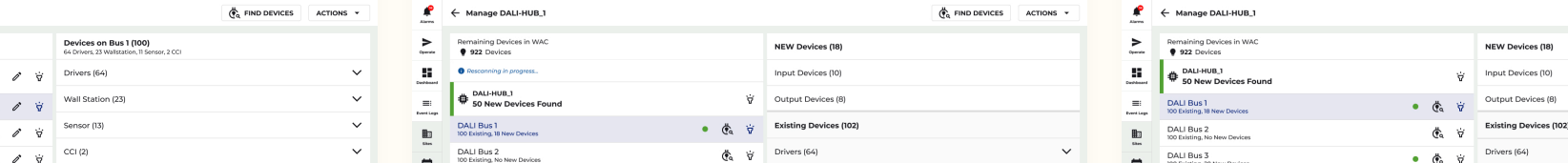
STEP 2 · NAME, HUB AND BUSES INLINE

STEP 3 · ADDED TO MAC, WITH REMAINING COUNT

THE PIVOT · V1 → V2

One modal, the *whole app behind it*

The actual V1 screen sat on top of a frozen interface for 10 to 27.5 minutes per hub. V2 deleted this blocking state entirely.



ONE CONTROLLER · WORST CASE DERIVED FROM THE STATED FIGURES (18 HUBS · 27.5 MIN)
GRAY · LOCKED · TEAL · WORKING · PALE · SCAN RUNNING BEHIND THE UI



THE REFRAME, IN ONE LINE

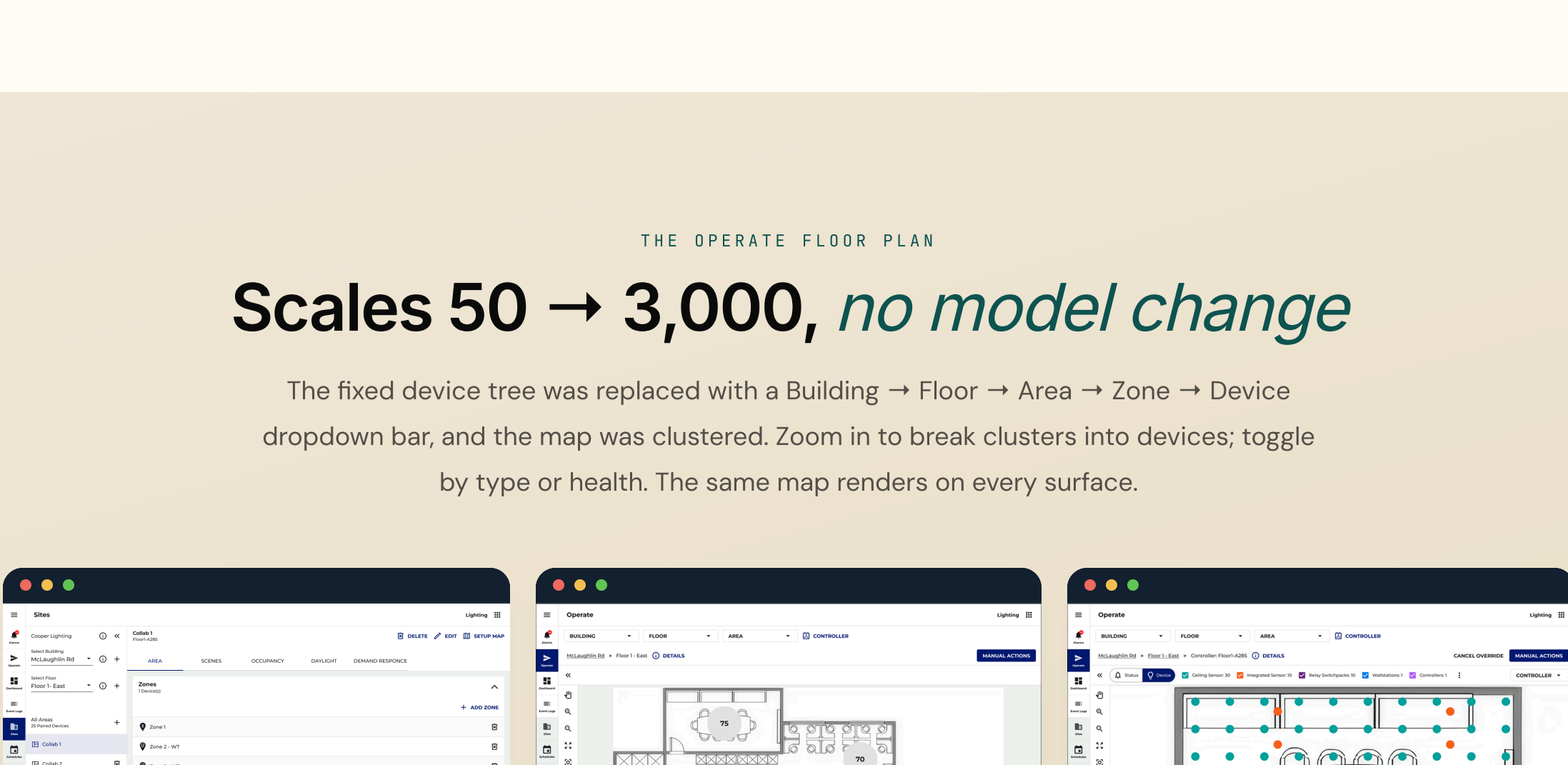
"We stopped designing for when the scan finishes, and started designing for what users can do while it's still running."

- Non-blocking: keep working during import.
- Two-banner progress model, always visible
- Hub-first modal matches MQTT pairing

LIFECYCLE

The full lifecycle, *after launch*

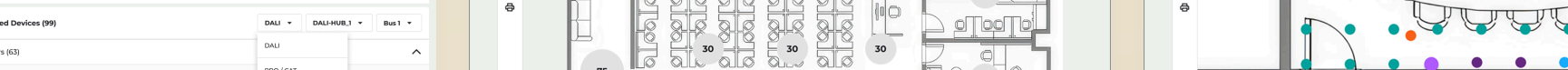
Commissioning is only the beginning. Hospitals expand, floors get reconfigured, devices get added to existing buses. Manage Hub supports the whole lifecycle, monitor health, rescans, and take corrective action without disrupting a live facility.



Manage Hub · Drivers 64, Wall Stations 23, Sensors 13, CCTs 2 per bus

Find Devices · a confirm dialog guards a live hub before any rescans

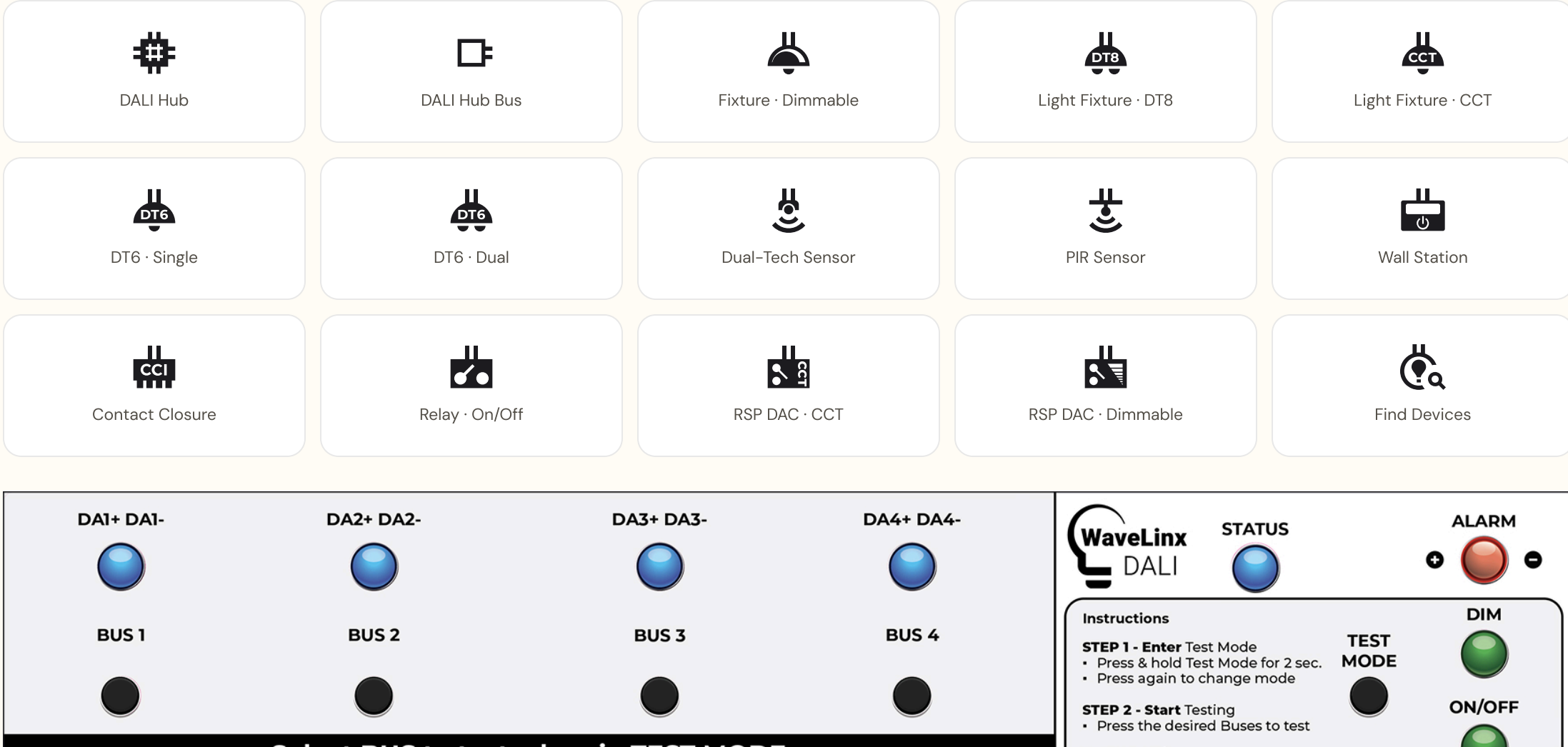
Add Devices · new vs existing, with a 5-8 min processing estimate



THE OPERATE FLOOR PLAN

Scales 50 → 3,000, *no model change*

The fixed device tree was replaced with a Building → Floor → Area → Zone → Device dropdown bar, and the map was clustered. Zoom in to break clusters into devices; toggle by type or health. The same map renders on every surface.

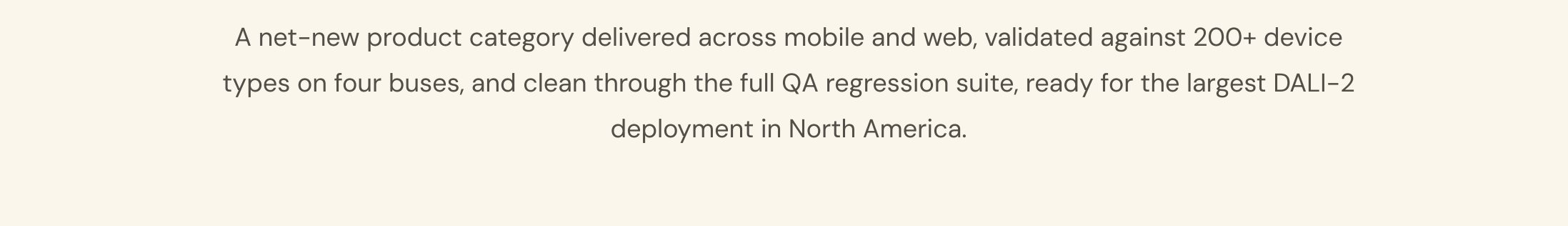


FILTER BEFORE LOAD · CLUSTERED AT FLOOR ZOOM · INDIVIDUAL DEVICES ON ZOOM-IN

CRAFT · A VISUAL LANGUAGE

The two-wire bus, in *every icon*

A custom-designed device icon set in Adobe Illustrator, embedding the DALI 2-wire bus, the defining physical trait of every DALI device, as a shared motif. Exported as SVG, validated for rendering via Fontello.



Press-ready 100x100mm hub label, die-cut and spec-complete with test-mode instructions, created in Illustrator and shipped on the hardware.

OUTCOME

Zero to one, *shipped and QA-clean*

A net-new product category delivered across mobile and web, validated against 200+ device types on four buses, and clean through the full QA regression suite, ready for the largest DALI-2 deployment in North America.

- Hub-first, non-blocking commissioning across the mobile app
- DTB dual-channel (CCT / white-tunable) driver support with auto-addressing
- Identify at hub, bus, and device level for physical-to-logical verification
- Custom DALI-2 icon system with a shared two-wire motif
- Filter-before-load area commissioning, scoped to a single 128-device bus
- Mixed-protocol zones: DALI, CAT wired, and WaveLinx wireless together

BY THE NUMBERS

Scope of the work

10,000+ Devices at full hospital scale

LEGACY UI 30-50
DALI-2 FLOOR 18,000+

12/12 QA regression scenarios passed

SCENARIOS 12 OF 12 PASSED

200+ Device types validated across 4 buses

COVERAGE 100% BUS 1 BUS 2 BUS 3 BUS 4

15 Device-type icons in the custom set

ICON SET 15 TYPES
SHOWN ON THE TWO-WIRE BUS MOTIF

COOPER LIGHTING · SIGNIFY · INDUSTRY-FIRST DALI-2 COMMISSIONING · FIRST RELEASE SHIPPED.



The engineering answer was a longer timeout. The UX answer was to make the entire wait invisible by designing around it.

THE ECHO

"DALI, Simplified." *the market said it back*

Cooper Lighting's own LinkedIn campaign now leads with the promise this work built around, automatic addressing, simplified wiring, commissioned like any WaveLinx system, with the shipped commissioning app as the campaign's hero image.

Marketing, sales, and product repeating the same simplification story is what it looks like when UX has orchestrated the outcome across every department.

NEXT CASE STUDY

Smart Lighting Mobile App →

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